Chapter 0 Part A – Class Expectations and Help in this Class

Calculus Physics II

Physics is a difficult subject for most students; it not only requires an understanding of math but more importantly a comprehension of the physics and the ability to connect the two though problem solving. This connecting the underlying principles in Physics with problem solving is a challenge for students and requires a different approach to studying than other subjects. Physics focuses on a few fundamental principles that are used to build an understanding of the physical world; therefore, a good grasp of these principles is a must. A student can attend all the lectures, read all the notes and watch hours of "Video" lectures but the only way to truly learn Physics is by doing it yourself and in small groups, practice, study, practice, study, practice, …! Learning Physics is a process, and this process needs to be learned and done over time, with patience and a commitment to learning.

Physics takes practice. You will not get the correct solution the first time, but don't give up, reassess and try again, just remember what Thomas Alva Edison said. **"I have not failed; I've just found 10,000 ways that won't work."**

How much time should I spend on this class?

- The standard rule of thumb is for each one credit hour in which you are enrolled, you should spend at least three hours working on this class's material per week. For a 15 week semester and 3 credit hour class, you should spend **9 hours/week** working on this class. This means at least 9 hours per week this semester on just Physics 2!
- This time can be spent watching lecture videos, reading lecture notes, attending office hours and review sessions, working on practice problems, attending lectures and meeting with your study group.
- You should put a schedule together now in order to succeed in this course. Tailor it to your own personal time commitments, but make sure you allow for all of the tasks that need to be done during this semester.
- Be realistic about your time when making your schedule each semester!

How should I spend my time?

- Conceptual material: Read the lecture notes and watch the lecture videos. You should develop an understanding of conceptual information by yourself and then with your study group.
- Applying your knowledge by attempting problems after you have reviewed lecture notes and videos.
- Work through the practice problems by yourself then with your study group. Review the solutions after you have attempted the problems. Learn from the worked-out solutions for the best practices in working out physics problems. Concentrate on the process of working out the problems, not just getting the correct answer.
- Asking for help or getting feedback from study group members, the instructor during office hours, during class time.

What are we doing during class time?

- The instructor will answer any questions about the material covered by this point.
- The instructor will review the more challenging material for students and work problems and demonstrate the best practices for problem solving.
- The instructor will conduct demonstrations to reinforce the physics being learned.
- There will be class activities that students will work on in small groups on the material covered that week.
- Announcements will be made about the class, activities or changes to the class.
- You will be able to connect with fellow students, get help, talk about challenges in the class and meet with your study group.

Create a STUDY GROUP: It is very beneficial in learning physics to find study partners and form a study group! Canvas will be set up with study groups under the "Peoples" in the "Groups" tab. If you have people in the class that you want to form a study group with you can self-assign yourself to a group during the first two weeks of class. After the second week I will randomly assign students to a group. You can also use your platform or app of choice to engage with your study group. **Why should you form a study group?**

1) Develop a better understanding of the material that is covered in the course.
2) Get new and different perspectives on the material.
3) Learn by teaching others.
4) Study more efficiently, quick help in getting un-stuck on problems so you don't waste time.
5) Gain a support group in the Physics class.
6) It forces you to study the material and be better prepared for the class assignments and exams.
7) Develop better problem-solving skills by practicing together.
8) It makes studying Physics more fun (if that is even possible!).
9) A study group helps you feel part of the class and less alone or isolated.
10) Improve your grade and get extra credit in the class.

Forming an effective study group can significantly enhance understanding and performance in this class. Here are some best practices to guide the process for putting together a group.

Choose the Right Members

- Aim for 3 to 5 students to keep the group manageable and focused.
- Try to include members with diverse strength, some may excel at math, others at conceptual understanding or explaining ideas.
- Choose peers who are committed, punctual, and respectful of others' time and effort.

Set Clear Goals and Expectations

- Define the purpose of the group: reviewing lectures, solving practice problems, preparing for exams, etc.
- Agree on ground rules for participation, preparation, and communication.
- Decide how often to meet (e.g., once or twice a week) and for how long.
- Set a location or method of gathering as a group. This could be a location in the UNF library, study room on campus, online (Zoom or other app) or off campus. Be flexible and be willing to change during the semester.

Prepare and Participate Actively

- Come to each group session prepared, having reviewed material or attempted problems beforehand.
- Rotate roles: someone can lead the session, another can take notes, and another can keep time.
- Encourage questions and explanations, teaching others is one of the best ways to learn.

Use Resources Wisely

- Bring lecture notes, practice problems, your notes and formula sheets to sessions.
- Use online simulations, videos, or problem-solving tools to clarify tough concepts.

Reflect and Adjust

- Periodically evaluate how the group is doing: Are meetings productive? Is everyone contributing?
- Be open to adjusting the format or schedule to better meet everyone's needs.

Study groups provide significant pedagogical benefits by fostering active engagement, collaborative problem-solving, and deeper conceptual understanding in an otherwise isolated environment. By working in structured groups, students must articulate their reasoning, defend their approaches, and reconcile differing interpretations of physical principles, which promotes conceptual clarity beyond individual study. These group also helps develop essential skills as students collectively work through multi-step problems rather than relying on passive solutions. Study groups further enhance accountability and motivation by creating a sense of community and shared responsibility, reducing the likelihood of disengagement. Group assignments can also mirror the collaborative nature of real world scientific work, reinforcing both the quantitative rigor and communication skills essential to success in physics.

What does "studying" really mean?

Many students feel like they are exerting effort and time studying for a particular course, but they perform poorly on graded assignments. The term "studying" in Physics means more than passively reading material, attending class, watching instructional videos, or reviewing problem solutions. Follow the guidance below to assist you in how to effectively take in information and process it in such a way that results in true learning.

BE AN ACTIVE LEARNER! - Read the Class Lecture notes, actively participate in the lecture, the class activities, study group, and do the practice problems. Keep up with all assignments and readings, do NOT wait to the last minute and seek help as soon as you need it. Don't hesitate to meet with your instructor or attend SASS help sessions.

ACTIVELY learn the course materials

- Take notes while you watch instructional videos. Pause the video to attempt an example problem yourself. Rewind to review key information. Merely watching/listening to a video or reading over a textbook will not result in true learning.
- Annotate your lecture notes. You can highlight and add notes to the conceptual reference material pages. Add notes when you are confused about a statement. Remind yourself to ask about that content during class, during study group time or in the instructor's office hours.
- Read the lecture notes posted on Canvas for each chapter before watching the lecture videos and before coming to class. Read the notes slowly and thoughtfully. This is not a novel; it is dense reading, take your time.
- If you still do not understand or need more clarification read the corresponding section in the online textbook "University Physics" in OpenStax.
- Attend class, come to class prepared, be engaged during class, ask questions. Make notes so you can ask about your questions when you are seeking help. Review the class notes and videos if needed.
- Form and participate in a study group, it is helpful to study alone and also to study in a group so that you verbalize and explain the ideas.

ASSESS your understanding

- Attempt questions and practice problems that have the solutions, this will give feedback as to whether you understood a concept or not. If you answer these questions incorrectly, go back and hunt for information to clarify your understanding.
- Ask for help from your study group or go to the instructor's office hours and ask for clarification on topics or specific questions. Your instructor wants to help you! They will be happy that you came to ask a question.
- Keep up in class, **don't get behind** in your reading or practice problems, Physics builds on previous material so the farther you get behind the problem is magnified!

ACTIVELY practice problem solving.

- Attempt problems that you have not worked through, this will require you to apply the physics in new ways and helps develop a deeper understanding.
- In-class worksheets are provided to guide you through using the information you just learned. You will work on these during scheduled class time in small groups to facilitate peer interaction and used as a study guide.
- Talk to your Study group and check your work with theirs. Ask them how they understood the concepts to see the material from a new perspective.
- Don't fall prey to "Virtual Understanding" – it makes complete sense when the instructor goes over it in class or video but as soon as you try to work a similar problem at home, it makes no sense. The best way to avoid this is to work out the examples that were done in the lecture or in the notes the same day and understand how they were solved, the way they were solved and why they were solved that way. Review notes right after class or watching the lecture when it is fresh in your memory. Re-Read section of the lecture notes that don't make sense.
- Work through the Practice problems, it is not required but it is the only way to develop your problem-solving skills, you must work problems and a lot of them.
- Work on good problem-solving techniques, don't just equation hunt!! Learn the concepts first before trying to solve the problems, understand the Physics before working problems. Use the "Top down" method of problem solving, start from the general principles, what type of physics is part of the problem, then think about what you know about this particular physics, then look at the specifics in the problem.
- Identify key concepts that are relevant to the problem statement. Review conceptual material related to those topics. Review your worksheets to see what you can apply to the problem.
- Take breaks often. Sometimes you need time to process the information before fully solving a problem. Come back to your work a few hours, or even a day later, and see what your fresh brain can do after digesting the information.
- It is alright to struggle with the material; this is what is expected. Using AI or online homework solutions like Chegg only hurt your ability to do well in this class!
- Read and use the "General Steps for Problem Solving in Physics" and "Problem Solving Check List" in Part B of Chapter 0.
- Read and use Rubric for grading worked out physics problems.

ASSESS your understanding

- Grade the accuracy of your work. Look at solutions and see how your work differs (if at all). Rework problems if you did not get them correct, review the process you used to solve the problem and make adjustments.
- Create exam-like conditions when preparing for an exam. Treat the practice problems and class activity worksheets as if it counted for points. Time yourself and do not look at your notes during the exam. Grade your work and revisit topics that you did not understand during the exam.

LEARN from your mistakes

- There are many levels of learning. It will take time and effort to climb the ladder to being an advanced learner. Do not be afraid of failures along the way. Making mistakes is the best way to learn!
- Don't give up. A low test score is not the end of the world. There will be more opportunities to make up for your mistakes and prove that you eventually mastered the topic.